Qiskit 2026 Track 1 — Three Qiskiteers

Code: `github.com/atharva7-g/qff` Slides: `slides/slides.pdf`

Part I: Exhaustive Search and Classification of Valid Passwords

The elliptic-curve background used here (isogenies, kernels, torsion subgroups, endomorphism rings, and supersingular curves) was studied using Andrew Sutherland's MIT 18.783 lecture notes [6]. The quantum-algorithmic background used below (Grover search, quantum walks, and query complexity) was studied using Ronald de Wolf's lecture notes [5]. The curve family and canonical bit conventions below are the assignment's prescribed ones.

Let

$$\mathbb{F} = \mathbb{F}_{1019}(i), \qquad i^2 = -1,$$

and let the initial curve parameter be

$$A_0 = 977 + 297i.$$

For a current parameter A_k, define the two possible transitions by

$$r_{0,1} = \frac{-A_k \pm \sqrt{A_k^2 - 4}}{2},$$

where the roots are ordered according to

$$\mathrm{key}(u + vi) = u + 1019v, \qquad 0 \le u, v < 1019.$$

The transition corresponding to bit $b \in \{0, 1\}$ is

$$\boxed{A_{k+1} = T_b(A_k) = \mathrm{canonical}(2 - 4r_b^2),}$$

where the canonical representative is the one of z and $-z$ having the smaller key.

For a 12-bit string

$$w = b_1 b_2 \cdots b_{12},$$

iterate these transitions to obtain

$$A_{12} = T_{b_{12}} \circ \cdots \circ T_{b_1}(A_0).$$

The corresponding endpoint is accepted precisely when

$$j(E_{A_{12}}) = \frac{256(A_{12}^2 - 3)^3}{A_{12}^2 - 4} = 797 + 921i \quad \text{in } \mathbb{F}.$$

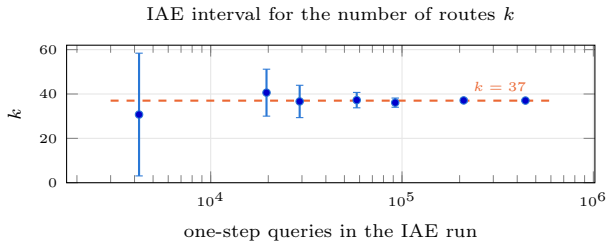
Exhaustive Enumeration. There are $2^{12} = 4096$ possible 12-bit strings. Evaluating the transition rule for each string and testing its terminal j-invariant gives exactly 37 valid passwords. These passwords fall into three classes according to their final canonical parameter. The full list of the 37 passwords is in `results/part1_L12.txt` of the repository.

Final parameter A_{12}	**Number of passwords**	**Terminal j-invariant**
$3 + 10i$	14	$797 + 921i$
$933 + 205i$	9	$797 + 921i$
$561 + 411i$	14	$797 + 921i$
Total	**37**	

Table 1: Classification of all valid 12-bit passwords.

Quantum completeness is a separate requirement. The list above is a finite-instance reference enumeration; classical enumeration alone is not the quantum completeness certificate required by the problem rules. In the unstructured phase-oracle model, finding all k marked routes among N possibilities has optimal query complexity $\Theta(\sqrt{Nk})$ [11], not the $\Theta(\sqrt{N/k})$ cost for finding just one. For $N = 4096$ and $k = 37$, the scale is $\sqrt{Nk} \approx 389$ phase-oracle queries. A complete submission must additionally specify how it handles unknown k, the case $k = 0$, and an overall failure probability ε; the classical list above does not establish those points. Each phase-oracle query must be charged for its compute–mark–uncompute implementation (24 one-step calls when $L = 12$).

Our quantum run. The one-step circuit is a 22-qubit permutation built from the survey (255 spots, 510 roads walked once). Iterative amplitude estimation on the phase oracle pins $k = 37$ at confidence ≥ 0.99 (Fig. 1, left); $k = 0$ would give an interval at 0 and the code reports "no route". Grover search with 8 iterations per run then returns all 37 routes, each confirmed by one classical walk. The run used 881,088 one-step queries, almost all of them in the high-confidence count (Fig. 1, right): at $N = 4096$ the constants of the counting dominate, while the scaling is $\sqrt{Nk}$.



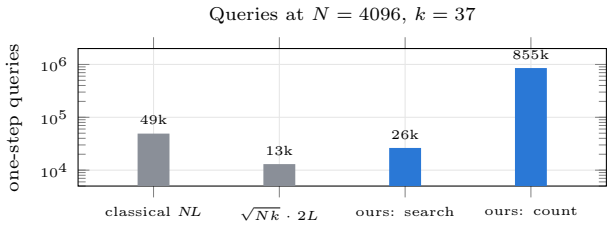


Figure 1: Part I. Left: the counting interval narrows to $k = 37$. Right: queries of our run against the classical and optimal quantum scales.

Part II: Quantum Search on the Supersingular Isogeny Graph

Target set and its density

Let V_p be the vertex set of the supersingular 2-isogeny graph over $\mathbb{F}_{p^2}$, and define the target set

$$T_p = \{j \in V_p : j \in \mathbb{F}_p\}.$$

Writing $j = u + vi$, with $u, v \in \mathbb{F}_p$ and $i^2 = -1$, the target condition is simply $v = 0$.
For $p \equiv 3 \pmod 4$, the Deuring–Eichler **Class-Number Formula** gives

$$|T_p| = \begin{cases} 2h(-p), & p \equiv 3 \pmod 8, \\ h(-p), & p \equiv 7 \pmod 8, \end{cases}$$

where $h(-p)$ is the class number of $\mathbb{Q}(\sqrt{-p})$ [3]. Moreover,

$$|V_p| = \frac{p}{12} + O(1) = \Theta(p).$$

By the Dirichlet class-number formula and Siegel's theorem,

$$h(-p) = p^{1/2+o(1)}.$$

Consequently, the proportion of target vertices is

$$\boxed{\frac{|T_p|}{|V_p|} = p^{-1/2+o(1)}.}$$

From vertex density to route density

A binary string $x \in \{0,1\}^L$ specifies a non-backtracking walk of length L. Let $A_L(x)$ denote its terminal curve parameter and define

$$f(x) = \begin{cases} 1, & j(E_{A_L(x)}) \in \mathbb{F}_p, \\ 0, & \text{otherwise.} \end{cases}$$

Let $q_L = \Pr_{x \leftarrow \{0,1\}^L}[f(x) = 1]$.
The supersingular 2-isogeny graph is 3-regular, and the Ramanujan property of supersingular isogeny graphs follows from the corresponding Hecke-operator eigenvalue bounds; see the survey of Florit Zacarías [7] and Pizer's Hecke-operator construction [9]. We use the non-backtracking random-walk estimate of Lubetzky–Peres [8, Theorem 3.5 and equation (3.13)]. A closely matching statement specialized to supersingular isogeny graphs is also given by Mula, Murru, and Pintore [10, Theorem 2.14]: a non-backtracking walk of length at least $(1+\epsilon)\log_\ell p$ has endpoint distribution within $\widetilde{O}(p^{-\epsilon/2})$ of uniform. For the standard supersingular 2-isogeny graph, connectedness is standard [9]; non-bipartiteness follows from the degree-two self-isogeny $[1+i]$ of the supersingular $j = 1728$ curve when $p \equiv 3 \pmod 4$ (so the graph has a loop, with loops retained). We also assume the two bits sample the two non-backtracking outgoing edges uniformly. Write $n_p = |V_p|$ and let $\widetilde{\mu}_L$ be the distribution of the final directed edge. For a fixed starting directed edge, their estimate gives, for $d = 3$ and $N_E = 3n_p$ directed edges,

$$\left\| \frac{\widetilde{\mu}_L}{\pi_E} - 1 \right\|^2_{L^2(\pi_E)} \le 2N_E 2^{-L}(8L^2 + 1),$$

where π_E is uniform on directed edges. By Cauchy–Schwarz,

$$d_{\mathrm{TV}}(\widetilde{\mu}_L, \pi_E) \le O\left(\sqrt{n_p}\, 2^{-L/2} L\right).$$

Projecting a directed edge to its endpoint cannot increase total variation distance, and the endpoint projection of π_E is uniform on vertices. Since $n_p = \Theta(p)$, taking

$$L = \lceil 3 \log_2 p \rceil$$

gives $d_{\mathrm{TV}}(\mu_L, U_p) = O(p^{-1} \log p) = \widetilde{O}(p^{-1})$. If the component is bipartite, the walk alternates between the two directed-edge classes; one must instead use the parity-conditioned stationary law and the target density in the reachable class. Under the stated non-bipartite hypothesis, total variation bounds probability discrepancy for every event, so

$$q_L = \frac{|T_p|}{|V_p|} + O(p^{-1} \log p) = \boxed{p^{-1/2+o(1)}}.$$

Thus a uniformly random length-L route is marked with probability q_L.

Grover search: construction of the marking oracle

There are $N = 2^L$ candidate routes and, writing k for the number of marked routes, $q_L = k/N$. Grover's algorithm searches this route space in superposition. The phase-oracle and amplitude-amplification viewpoint is standard; see de Wolf's quantum-computing notes [5]. Starting with

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^L} |x\rangle,$$

a reversible computation evaluates the L isogeny steps controlled by x, computes the terminal invariant

$$j(E_{A_L(x)}) = u(x) + v(x)i,$$

and sets a flag equal to $[v(x) = 0]$. A phase flip is applied when this flag is 1. The computation is then uncomputed, so the route register is left unchanged except for its phase. The resulting phase oracle is

$$\boxed{O_f|x\rangle = (-1)^{f(x)}|x\rangle.}$$

In particular, this is not an assumed free oracle: it is implemented by L forward isogeny-step computations, the reversible target test, and L inverse steps. Thus one marking operation costs $2L$ one-step calls, apart from the finite-field arithmetic for the test.
The Grover diffusion operator $D = 2|s\rangle\langle s| - I$ is then alternated with O_f. If

$$\theta = \arcsin \sqrt{q_L},$$

after t iterations the success probability is

$$P_G(t) = \sin^2((2t+1)\theta).$$

Choosing $t = \Theta(q_L^{-1/2})$ gives constant success probability. Hence

$$t = O(p^{1/4+o(1)}), \qquad \boxed{Q_{\text{step}} = p^{1/4+o(1)}.}$$

This is the quantum upper bound, including the $L = \Theta(\log p)$ cost of each marking operation.

Classical comparison

Independent random route evaluations succeed with probability q_L. The expected number of evaluations before the first success is $1/q_L$. Therefore

$$Q_{\text{classical}} = \Theta(q_L^{-1}) = p^{1/2+o(1)}$$

complete-route evaluations, or

$$\boxed{Q_{\text{classical,step}} = p^{1/2+o(1)}}$$

one-step calls. Grover's algorithm gives a quadratic improvement in the inverse success probability.

Lower bound in the black-box model

To state a rigorous lower bound, consider the **Unstructured Quantum Search Model**. Its input is an unknown Boolean function

$$f : X \longrightarrow \{0,1\}, \qquad X = \{0,1\}^L, \quad |X| = N,$$

promised to have exactly k marked inputs. The algorithm accesses the marked set only through the oracle

$$O_f|x,z\rangle = |x, z \oplus f(x)\rangle$$

and must output a marked x.
The **Zalka Optimality Theorem** [1] and the BBBV lower bound [4] state that any quantum algorithm achieving constant success probability requires

$$\Omega\left(\sqrt{\frac{N}{k}}\right) = \Omega(q_L^{-1/2})$$

oracle queries. Since $q_L = p^{-1/2+o(1)}$, this becomes

$$\boxed{Q_{\text{oracle}} = \Omega(p^{1/4-o(1)}).}$$

Together with Grover's upper bound, this establishes the $p^{1/4+o(1)}$ search exponent in the unstructured black-box model.
The model qualification is essential. The actual isogeny transition is explicit algebraic finite-field arithmetic. An algorithm with direct access to that structure may exploit relations between routes or vertices that an arbitrary Boolean oracle does not reveal. Thus Zalka's theorem is a lower bound for the black-box abstraction, not an unconditional lower bound for every algorithm solving the explicit arithmetic problem. If each route-oracle query is required to be implemented serially using $2L$ step calls, the corresponding step-call lower bound in that restricted model is $\Omega(Lp^{1/4-o(1)})$.
Related path-finding algorithms. The standard ℓ-IsogenyPath problem asks, given two supersingular curves, for any path between them; ℓ fixes the degree of each edge, not the number of edges [12]. Page and Wesolowski [13, Proposition 8.6] give a generic bidirectional algorithm with expected time $\text{poly}(\ell, \log p)p^{1/2}$, memory $O(p^{1/2})$, and output path length $O(\log p)$. For fixed $\ell = 2$, this is $\widetilde{O}(\sqrt{p})$. In the present setting one may choose a fixed Clock-Tower target: $j = 1728$ if $p \equiv 7 \pmod{12}$, or $j = 0$ if $p \equiv 11 \pmod{12}$. The algorithm is therefore a relevant classical benchmark, but it uses walks from both endpoints and a collision search, so it is explicitly forbidden by the assignment's no-backtracking/no-meet-in- the-middle rule.
Delgado [14] gives an unconditional Las Vegas algorithm with expected time and memory $p^{1/3}\exp(O(\sqrt{\log p \log\log p})) = p^{1/3+o(1)}$ for OneEnd; known reductions extend this to EndRing and unrestricted isogeny problems. This is not directly an unconditional fixed-ℓ path algorithm. Transferring it to ℓ-IsogenyPath through Wesolowski's reduction [12] invokes GRH. Moreover, that reduction uses a path parameter $e = \lceil (\log p)^c / \log \ell \rceil$, giving a polylogarithmic length bound, not automatically a $\Theta(\log p)$ bound. These results reinforce that the black-box lower bound here is not a lower bound for every algorithm using the explicit arithmetic structure.

Experimental exponent fitting

For each tested prime p, let M_p be the number of random routes sampled and S_p the number ending in $\mathbb{F}_p$. The empirical density is

$$\widehat{q}_p = \frac{S_p}{M_p}.$$

To test the predicted power law, fit

$$\widehat{q}_p \approx C p^{-\alpha}, \qquad \log \widehat{q}_p = \log C - \alpha \log p.$$

For 65 primes $p \equiv 3 \pmod 4$ from 2^{10} to 2^{26}, with $4 \cdot 10^5$ random routes of length $L = 2\lceil \log_2 p \rceil$ each, the reported regression is

$$\boxed{\widehat{q}_p \approx 8.7\, p^{-0.523},} \qquad \boxed{\widehat{\alpha} = 0.523 \pm 0.040,}$$

and the exact spot fractions (40 primes up to 2^{20}) give $\widehat{\alpha} = 0.560 \pm 0.081$ (two standard errors). This is consistent with the theoretical exponent $\alpha = \frac{1}{2}$; finite-sample variation and class-number fluctuations mean that the fitted exponent need not equal $\frac{1}{2}$ exactly. The experiment is supporting evidence, not a proof of the asymptotic estimate.

Demonstration. On the campus $p = 5923$, $A_0 = 66 + 1530i$, $L = 12$, a fraction $q = 1/39.8 \le 1/32$ of the routes ends on the Clock Tower. The 25-qubit circuit (transpiled to $\{cx, rz, sx, x\}$, one Grover iteration: 24 queries, $3.40 \cdot 10^6$ CX, depth $1.11 \cdot 10^7$) follows $\sin^2((2t+1)\theta)$ to four decimals and reaches 0.981 at $t = 4$, where classical walking with the same number of queries reaches 0.205 (Fig. 2, right). With q unknown, BBHT needs on average 91 queries and 7.9 classical checks, against 477 steps for classical random routes.

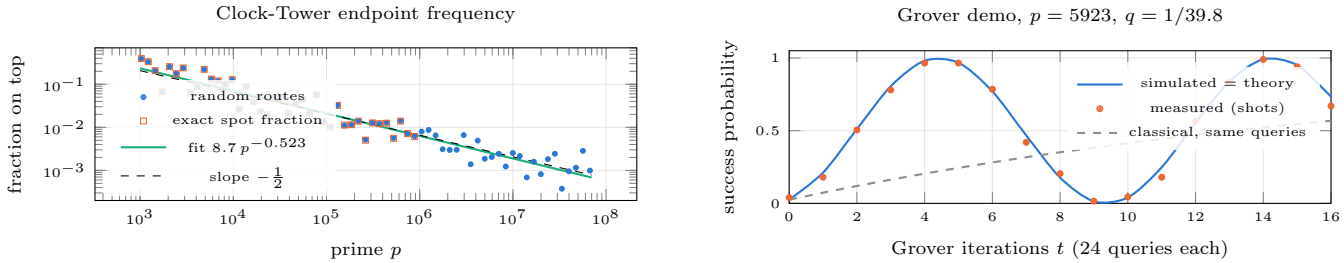


Figure 2: Part II. Left: fraction of routes ending on the Clock Tower against p. Right: Grover success against iterations, with the theory curve $\sin^2((2t+1)\theta)$ and classical random routes with the same number of queries.

Part III: Reversible Isogeny Arithmetic and Resource Estimate

Field registers. Put $n = \lceil \log_2 p \rceil$. Encode $z = a + bi \in \mathbb{F}_{p^2}$ by two n-bit residues (a, b), with $i^2 = -1$. Thus one curve parameter takes $2n$ qubits. Addition is componentwise modulo p, and

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i \pmod p.$$

A nonzero inverse is $(a - bi)/(a^2 + b^2)$, since $a^2 + b^2 \neq 0$ for nonzero $a + bi$ when $p \equiv 3 \pmod 4$. Modular addition, multiplication, comparison, and multiplication by fixed constants therefore give reversible arithmetic circuits over the field. Our blocks use only X, CX and Toffoli gates: Qiskit's CDKM ripple-carry adder [15] on $n + 1$ bits, a modular adder (add, subtract p, add p back under the sign flag, clear the flag with a MAJ-ladder comparator of $2n$ Toffolis), an in-place doubling (a relabelling of qubits), a multiply-accumulate $z \mathrel{+}= xy$ by shift-and-add, and Karatsuba's three products for $\mathbb{F}_{p^2}$.

Square root and deterministic branch selection. For $D = A^2 - 4 = \alpha + \beta i$, the assignment's supersingularity/full-2-torsion promise ensures that D is a square in $\mathbb{F}_{p^2}$. A useful explicit square-root reduction is as follows. Compute $m = \alpha^2 + \beta^2 \in \mathbb{F}_p$ and a base-field square root s of m. Of $s, -s$, select the one for which

$$u = (s + \alpha)/2, \qquad v = (s - \alpha)/2$$

are squares (zero is allowed). If $D = (x + yi)^2$, this choice is $s = x^2 + y^2$, and hence $u = x^2$, $v = y^2$. Because $p \equiv 3 \pmod 4$, a nonzero square $t \in \mathbb{F}_p$ has square root $t^{(p+1)/4}$; zero is handled separately. The circuit needs only two base-field exponentiations: $\lambda = m^{(p+1)/4}$, $\delta = (\alpha + \lambda)/2$ (replaced by α if $\delta = 0$, which happens only for $\beta = 0$), and $r = \delta^{(p-3)/4}$. Then $y = r\delta$, $t = \beta r/2$ and the flag $q = [yr = 1]$ give $\sqrt{D} = y + ti$ if $q = 1$ and $\sqrt{D} = -t + yi$ if $q = 0$, because $1/y = \pm r$; no inversion and no second square root are needed. Finally compare the keys of $r_+ = (-A + \sqrt{D})/2$ and $r_- = (-A - \sqrt{D})/2$ and sort them. This last comparison, not the arbitrary sign returned by a square-root routine, fixes the route-bit convention.

The transition circuit then computes

$$D = A^2 - 4, \quad r_0, r_1 = \frac{-A \pm \sqrt{D}}{2}, \quad A' = \mathrm{can}(2 - 4r_b^2),$$

where $\mathrm{can}(z)$ compares $\mathrm{key}(z)$ and $\mathrm{key}(-z)$. The output formula is the stipulated parameter transition; no coordinate change or extra square root for model normalization is needed in the circuit. All comparisons and branch choices are reversible controlled operations (flags and controlled swaps).

Reversible embedding. The map $(A, b) \mapsto T_b(A)$ is not assumed injective. It must therefore not be implemented by erasing A. Instead implement the XOR/compute form

$$|A\rangle|b\rangle|0\rangle \longrightarrow |A\rangle|b\rangle|T_b(A)\rangle,$$

retaining the inputs and uncomputing the square-root and arithmetic work registers (Bennett: compute, copy A', run the computation backwards). For a route $x = b_1 \cdots b_L$, compute successive $A_1, \ldots, A_L$ in scratch registers, evaluate the endpoint predicate, apply a phase $(-1)^{f(x)}$, and run the arithmetic circuit backwards. The route register is untouched. This is a valid phase oracle even though an individual transition is not necessarily a permutation. A direct implementation stores all intermediate curve registers; reversible pebbling can trade fewer registers for recomputation. We keep the direct implementation, which minimises the gate count.

For the Clock-Tower test, write

$$N = 256(A^2 - 3)^3, \qquad D = A^2 - 4, \qquad j = N/D.$$

Since $D \neq 0$ and Frobenius acts by $(a+bi)^p = a - bi$,

$$j \in \mathbb{F}_p \iff j^p = j \iff \overline{N}D = N\overline{D}.$$

Thus the marker can be computed by a field multiplication and a zero test on the two coordinates of $\overline{N}D - N\overline{D}$, without a field inversion. This is preferable to explicitly dividing to obtain j.

Validation at $p = 1019$. Every gate is X, CX, CCX or CSWAP, so the circuit maps basis states to basis states and Aer's matrix-product-state simulator runs the 587-qubit circuit with bond dimension 1. For every canonical spot and both bits (510 cases) the measured output equals an independent classical implementation of $T_b(A)$, the inputs are unchanged and every work qubit is 0; the Clock-Tower test is correct on all 255 spots. A permutation matrix that is correct on every basis state is correct on every superposition, by linearity. As a further check, 64 states $\cos w|x\rangle + e^{i\phi} \sin w|y\rangle$ pass through the new circuit and then the inverse of a reference circuit and return to $|0\rangle$, which fixes all relative phases (a deliberate phase error is detected); 32 cases on random spots at $p = 46351$ also pass.

Measured gate accounting. Counts are exact, in the basis $\{x, cx, ccx\}$ (a CSWAP is one Toffoli and two CX), from a memoised walk over the block hierarchy, so no circuit is ever built gate by gate at large p. In the Qiskit basis $\{cx, rz, sx, x\}$ every Toffoli becomes 6 CX (a CSWAP 8), so the two-qubit count is CX $+ 6 \cdot$ Toffoli. Table 2 shows how each design choice changes the step; the square root was 95% of the baseline Toffolis (Fig. 3, middle). Under a standard exact CCX decomposition, G Toffolis contribute $7G$ T gates ($4G$ with one measured ancilla per Toffoli).

step circuit at $p = 1019$	qubits	Toffoli	CX	2-qubit in $\{cx, rz, sx, x\}$	depth
baseline (Algorithm 9 root [16], subtract-and-add-back flag)	850	431,914	900,472	3,491,956	1,036,521
+ comparator	850	380,506	797,656	3,080,692	891,043
+ Karatsuba	850	339,162	715,608	2,750,580	797,033
+ norm square root	561	139,236	291,748	1,127,164	325,241
+ parallel squaring (submitted)	587	139,236	291,748	1,127,164	307,933
Clock-Tower test (submitted)	192	37,794	–	–	63,148

Table 2: Part III. Cost of one step at $p = 1019$; each row adds one change.

Cost at $p \simeq 2^{256}$. The exact counts for primes of $n = 10$ to 256 bits fit Toffoli $\approx 127\, n^{2.97}$ and qubits $\approx 4.4\, n^{1.94}$ (Fig. 3, left), matching the $O(n^3)$ bound of a schoolbook multiplier inside $O(n)$-step exponentiations. At $n = 256$ one step costs $1.83 \cdot 10^9$ Toffolis (0.34 of the baseline) on 204,049 qubits, and the Clock-Tower test $2.1 \cdot 10^7$ Toffolis. For the search we take $L = 2n = 512$: the walk data of Part II show that $L = 2\lceil \log_2 p \rceil$ already gives a route fraction equal to the spot fraction (the proof above uses $3 \log_2 p$ for a clean bound; that choice multiplies the cost by 3/2). With $q \simeq 7/\sqrt{p} = 2^{-124.9}$ (the median of $q\sqrt{p}$ in our data), Grover needs $(\pi/4)/\sqrt{q} \simeq 2^{62.1}$ iterations (at most 2.86× more on average with BBHT if q is unknown [17]). Each iteration is $2L$ steps, one Clock-Tower test and the diffusion, $1.88 \cdot 10^{12}$ Toffolis, so the search costs about $2^{102.9}$ Toffolis ($6.6 \cdot 10^{31}$ T gates), $2^{105.9}$ two-qubit gates in $\{cx, rz, sx, x\}$, depth $2^{104.4}$, on about 466,000 logical qubits. Classical random routes require about $2^{124.9}$ route evaluations, or about $2^{133.9}$ one-step evaluations. Under a depth limit of 2^m (NIST MAXDEPTH) the search splits over $(2^{104.4-m})^2$ machines and the total grows by $2^{104.4-m}$ (Fig. 3, right): $2^{111.3}$ Toffolis for $m = 96$. These figures are logical counts, not a fault-tolerant hardware estimate.

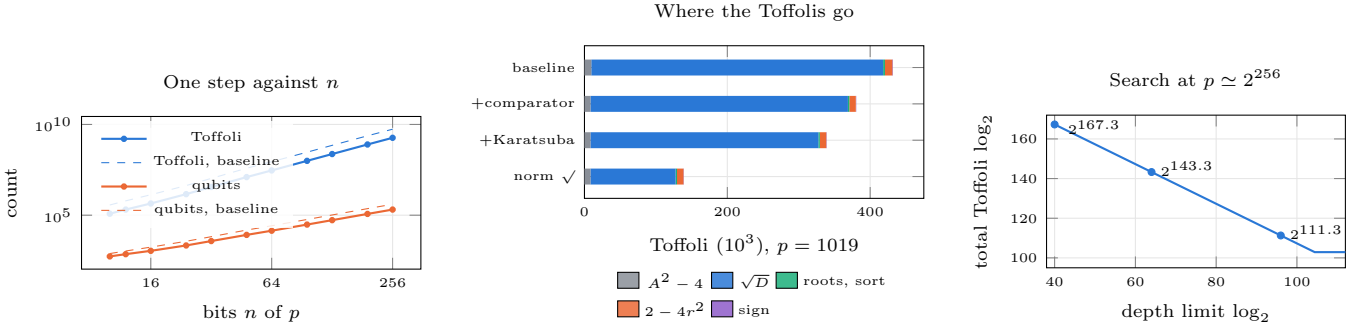


Figure 3: Part III. Left: exact cost of one step for primes of n bits (solid: submitted circuit; dashed: baseline). Middle: Toffolis per stage of the step. Right: total cost of the Part II search under a depth limit.

Part IV: Randomly Relabelled (Djinn) Oracle Model

Let V be the graph's vertex set, $|V| = N$, and let $\pi : V \to [N]$ be a uniformly random hidden bijection. A vertex name alone is not in general a sufficient routing state: the two legal forward roads depend on the road used to enter the current vertex. Let $e = (u, v)$ denote the directed arrival edge (or use an incoming-port label if parallel edges are present). If the two non-backtracking successors of e are $e_b = (v, w_b)$, define the relabelled successor by

$$\mathcal{N}_b(\pi(u), \pi(v)) = \pi(w_b), \qquad b \in \{0, 1\}.$$

The coherent neighbour oracle must therefore retain the arrival-edge data, for example

$$O_{\mathcal{N}} : |x_{\text{prev}}, x_{\text{cur}}, b, z\rangle \mapsto |x_{\text{prev}}, x_{\text{cur}}, b, z \oplus \mathcal{N}_b(x_{\text{prev}}, x_{\text{cur}})\rangle.$$

The volunteer predicates depend only on the current name, so their oracle may be written

$$O_f : |x_{\text{cur}}, t\rangle \mapsto |x_{\text{cur}}, t \oplus f^{\pi}(x_{\text{cur}})\rangle.$$

All input registers are preserved, making these XOR interfaces unitary even when the successor map is not one-to-one. A vertex-only oracle $\mathcal{N}_b(x)$ describes the original non-backtracking routing rule only if the problem separately defines road labels independently of the incoming road; otherwise the incoming edge/port must be part of the state. The initial state must likewise specify the prescribed starting port or incoming edge.

What survives relabelling? A permutation changes names, not the underlying graph. Degree, adjacency, connectedness, cycles, expansion/spectrum, and the size and location of the marked set as a subset of vertices are preserved. Classical random walks, breadth-first or depth-first exploration with a visited-name table, and collision/cycle detection therefore remain available. If the Clock-Tower density is $\tau = |T_p|/|V| = p^{-1/2+o(1)}$ and the walk is sufficiently mixed, classical random sampling needs $\Theta(1/\tau) = p^{1/2+o(1)}$ marker tests; exhaustive graph exploration costs $O(N)$ neighbour queries and $O(N)$ stored names. Arithmetic operations on A, direct j-invariant evaluation, Frobenius calculations, and the quaternion/class-group methods are not directly available from opaque names alone.

If the neighbour and marker oracles are genuinely coherent, amplitude amplification over valid directed-edge states or route strings still gives a quadratic search speedup, provided the relevant uniform starting state and inverse oracle are available. Moreover, if a reversible walk on the underlying graph (or a suitable reversible lift) can be accessed coherently, its constant spectral gap can support a MNRS/Szegedy-type quantum-walk search [2,5] with cost

$$S + O\left(\frac{1}{\sqrt{\tau}}\left(\frac{1}{\sqrt{\delta}}U + C\right)\right),$$

where S is setup, U is a coherent walk update, C is the coherent marker check, and δ is the Markov-chain spectral gap. For a constant-gap expander and comparable U, C, this is again of order $p^{1/4+o(1)}$ oracle operations. Random naming does not destroy expansion. However, ordinary classical neighbour responses or human volunteers are not coherent oracles: they cannot be queried on a superposition, so Grover and quantum-walk search do not apply to that interface without an efficient reversible implementation of the responses.

Searching for both predicates. If success requires both Clock Tower and ANC, the relevant marked density is

$$\tau_\cap = \frac{|T_p \cap S_{\mathrm{ANC}}|}{|V|},$$

not automatically $\tau_{\mathrm{CT}}\tau_{\mathrm{ANC}}$. No independence assumption is justified by random relabelling. Conditional densities or an explicit intersection count are needed before assigning a complexity.

Lower-bound qualification. Zalka/BBBV search lower bounds [1,4] give $\Omega(1/\sqrt{\tau})$ quantum marker queries when the marked set is an otherwise unstructured unknown subset and auxiliary oracles reveal no information correlated with it. A random permutation of names alone does not establish that promise: the graph still has exploitable topology, and the true Clock-Tower set is arithmetically correlated with the graph. Therefore the rigorous conclusion is conditional: the $p^{1/4-o(1)}$ lower bound applies to the explicitly unstructured marker-oracle abstraction, not automatically to the full Djinn graph-oracle problem. To claim a lower bound for Part IV, the oracle family must specify whether the marked set is random independently of the graph, whether coherent access is allowed, and what the neighbour oracle reveals.

References

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